

Wireless Networks



Tutorial 1

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By Pushkar Maurya

Network Card, Wireshark & Packet Tracing

Structure of Tutorial

1. Know Your Wireless Network
2. Modes of Operation of Wireless Adapter
3. Using Wireshark
4. Packet Tracing Through Managed Mode
5. Monitor Mode
6. Enabling Monitor Mode
7. Resources

Know Your Wireless Network

> sudo ifconfig

```
enp0s20f3: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 10.0.3.1 netmask 255.255.255.0 broadcast 10.0.3.255
    ether 00:16:3e:00:00:00 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<LOOPBACK,UP, RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 1 scopeid 0x20<link>
    ether 00:00:00:00:00:00 txqueuelen 1000 (Ethernet)
    RX packets 28677 bytes 21910203 (21.9 MB)
    RX errors 0 dropped 2 overruns 0 frame 0
    TX packets 5444 bytes 1959469 (1.9 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

> sudo iwconfig

```
lo: IEEE 802.11 ESSID:"LAPTOP-S"
    Mode:Managed Frequency:5.22 GHz Access Point: 30:86:2D:77:81:20
    Bit Rate=573.5 Mb/s Tx-Power=22 dBm
    Retry short limit:7 RTS thr:off Fragment thr:off
    Encryption key:off
    Power Management:on
    Link Quality=67/70 Signal level=-43 dBm
```

Modes of Operation of Wireless Adapter

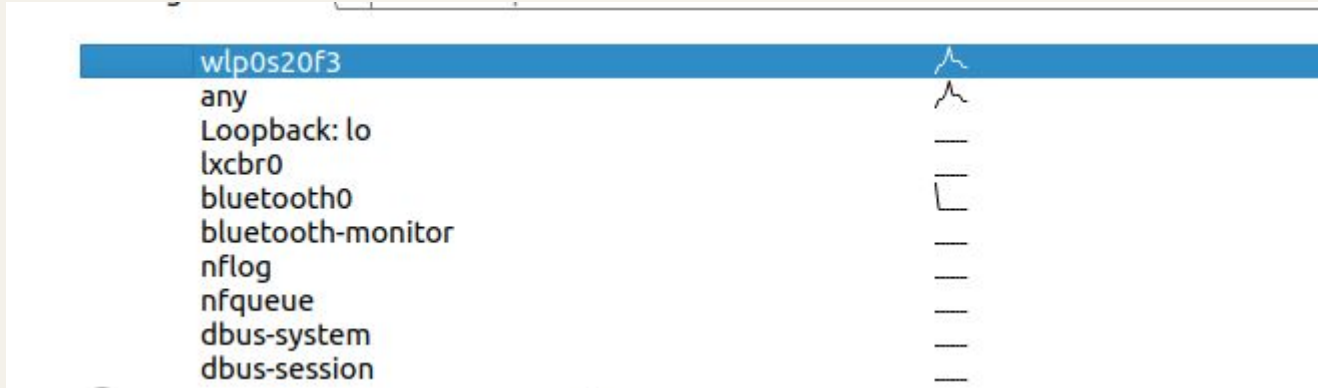
- Master (Access Point)
- Managed (Client/Station) (Default)
- Ad hoc
- Repeater
- Mesh
- WiFi Direct (P2P)
- TDLS (Tunneled Direct Link Setup)
- Monitor Mode

› **sudo iw list**

```
Supported interface modes:  
* IBSS  
* managed  
* AP  
* AP/VLAN  
* monitor  
* mesh point  
* P2P-client  
* P2P-GO
```

Using Wireshark

- Installing Wireshark: You can use any of the following
 - Use
 - **sudo apt install wireshark**
 - Or download .deb package from here (<https://www.wireshark.org/download.html>)
- Starting
 - **sudo wireshark**



Packet Tracing Through Managed Mode

1. Select Interface
2. Packet Types
3. Packet Selection
4. Apply as Filter

L7	Apply as Column	Ctrl+Shift+I	MDNS	554 ✓
L7	Apply as Filter		Apply as Filter: udp	534 ✓
L7	Prepare as Filter		Selected	453 ✓
L7	Conversation Filter		Not Selected	473 ✓
L7				496 ✓

5. Apply as Column

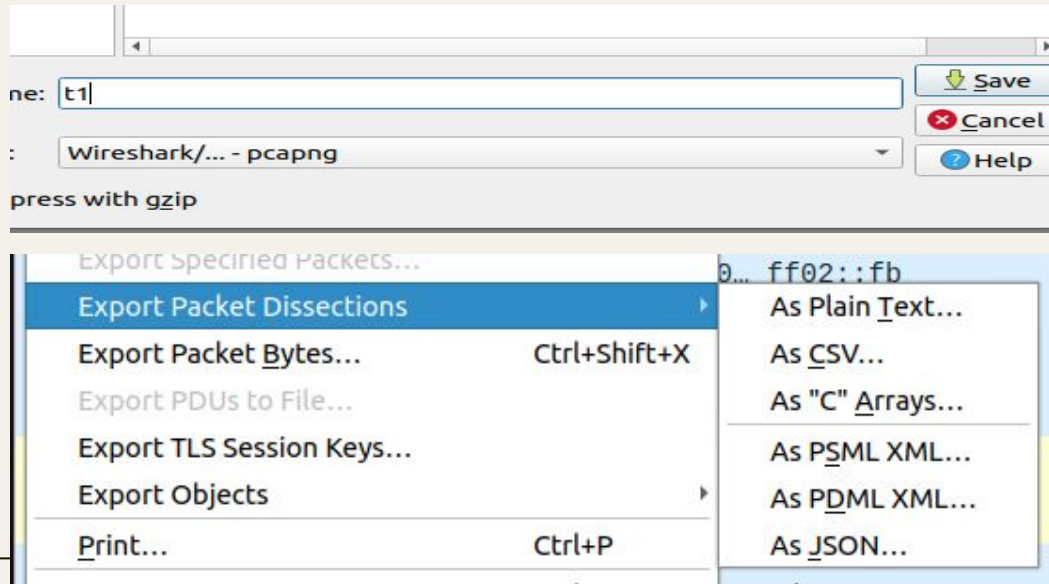
1. 928989680	Apply as Column	Ctrl+Shift+I	DNS	78
1. 928989730	Apply as Filter		DNS	76
1. 928989782			BNS	9

Packet Tracing Through Managed Mode

6. Export as Pcapng/Pcap

7. Export as CSV/Json

8. Run Python on Exported File



Monitor Mode

Why monitor mode?

- It allows to capture packets in the air of all reachable APs and STAs without connecting to any access point
- Help design better WiFi network
- Help test security of WiFi to avoid attacks like MITM
- And much more

Enabling Monitor Mode

Before :

```
(base) pushkar@pushkar-IdeaPad-3-15ITL6:~$ sudo iwconfig
[sudo] password for pushkar:
lo          no wireless extensions.

wlp0s20f3   IEEE 802.11  ESSID:"LAPTOP-S"
Mode:Managed  Frequency:5.22 GHz  Access Point: 30:86:2D:77:81:20
Bit Rate=390 Mb/s   Tx-Power=22 dBm
Retry short limit:7  RTS thr:off   Fragment thr:off
Encryption key:off
Power Management:on
Link Quality=56/70  Signal level=-54 dBm
Rx invalid nwid:0  Rx invalid crypt:0  Rx invalid frag:0
Tx excessive retries:0 Invalid misc:31 Missed beacon:0

lxcbr0      no wireless extensions.

(base) pushkar@pushkar-IdeaPad-3-15ITL6:~$
```

*Note Down Your Interface

```
wlp0s20f3 IEEE 802.11
Retry short li
```

Enabling Monitor Mode

Steps :

```
(base) pushkar@pushkar-IdeaPad-3-15ITL6:~$ sudo iwconfig wlp0s20f3 mode monitor  
lo                no wireless extension  
wlp0s20f3 IEEE 802.11  
                Retry short limit 7  
                Power Management disabled
```

```
e) pushkar@pushkar-IdeaPad-3-15ITL6:~$ sudo ifconfig wlp0s20f3 down  
e) pushkar@pushkar-IdeaPad-3-15ITL6:~$ sudo iwconfig wlp0s20f3 mode monitor  
e) pushkar@pushkar-IdeaPad-3-15ITL6:~$ sudo ifconfig wlp0s20f3 up  
e) pushkar@pushkar-IdeaPad-3-15ITL6:~$
```

Enabling Monitor Mode

After :

```
(base) pushkar@pushkar-IdeaPad-3-15ITL6:~$ sudo iwconfig  
lo          no wireless extensions.  
  
wlp0s20f3   IEEE 802.11  Mode:Monitor  Tx-Power=-2147483648 dBm  
            Retry short limit:7   RTS thr:off   Fragment thr:off  
            Power Management:on  
  
lxcbr0      no wireless extensions.  
  
(base) pushkar@pushkar-IdeaPad-3-15ITL6:~$
```

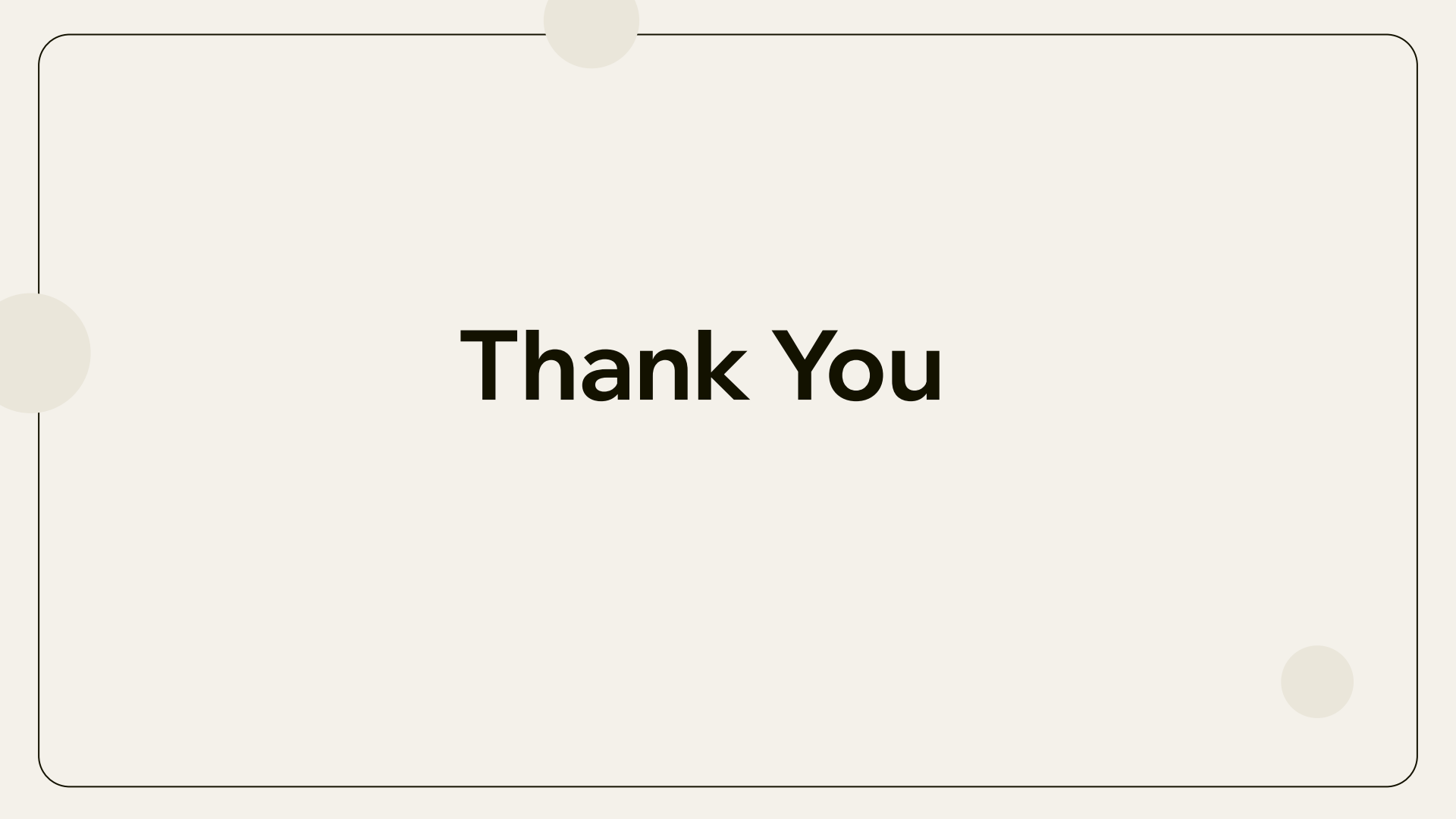
*Same as Managed Mode ,Everything Can be done in Monitor Mode

*More Packets are Captured In Monitor Mode ,better analysis

Resources

<https://www.youtube.com/watch?v=lb1Dw0elw0Q>

<https://www.youtube.com/watch?v=5zHssSXp2sM>



Thank You